Which Stride Broke Down – and When?

Fatigue vs bad biomechanics in ice hockey · Hugo Persson & Thrainn Leo Thorisson · MSc Sports Technology, KTH

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Summary

We propose a system that shows a coach which skating movement breaks down, and when, so that fatigue-related changes in technique can be caught before they turn into groin, hip or knee injuries. Positioning sensors in the arena roof (RTLS) record *what* a player does, split into skating actions such as forward strides, crossovers and stops. IMUs on the skates record *how* each stride is performed. Fusing the two labels every stride with its action and compares it with the player's own rested baseline.

The project is a concept based on desktop research. The coach view shown on our poster uses simulated data; validating the system on real ice is the next step.

Background and problem

Groin and hip injuries are the biggest injury problem in ice hockey. Half of European professional players and one in five NCAA players report them as the dominant problem [1]. Strength imbalance and reduced hip range of motion are named risk factors, often linked to poor skating mechanics [1]. In one NHL team, players whose hip adductor strength was below 80% of their abductor strength were 17 times more likely to suffer an adductor strain [2].

Fatigue adds to the risk. Hockey is played in short, high-intensity shifts, and skating technique changes as players tire. Wearable sensors on skaters have shown that joint angles fluctuate more, the legs move less in sync and the trunk position shifts with fatigue [3]. Skating movements other than forward striding, such as crossovers, turns and stops, are barely studied at the hip [1].

Current technology does not show *how* a player skates during a game:

- **Commercial wearables** focus on speed, acceleration, load and return-to-skate progression [4].
- **Video and markerless motion capture** match marker-based systems for the lower limbs, but errors grow in fast movements and hip angles are the least accurate [5]. Bulky pads also hide the joints, so cameras track the equipment rather than the body.

- **Body-worn sensor networks** with IMUs on the thigh, shank and pelvis shift under protective gear during collisions and falls.
- **Skate-mounted IMUs** are accurate for stride time, contact time and stride length, but have only been validated in a lab: on a 14 m synthetic-ice track, forward skating only, at up to 2.63 m/s [6].

We know hip imbalance raises injury risk, and that fatigue changes skating mechanics. No study has yet linked the two on real ice. That is the gap this project targets.

Aim and research question

The aim is to design a system that monitors skating biomechanics during real games and tells the coach when, and in which movement, a player's technique starts to break down.

Research question: Can positioning data and skate-mounted IMUs together detect which skating action deviates from a player's own baseline, and separate fatigue-driven changes from persistent poor technique?

Proposed system

The system has two sensing layers whose data are fused stride by stride:

1. **RTLS: what the player does.** Receivers mounted in the arena roof track a tag in each player's shoulder pads. Swarén et al. synchronised positioning data with game video from the Swedish Hockey League and identified nine skating characteristics, including forward and backward skating, crossovers left and right, glides, C-cuts, sharp turns and stops [7]. We use the same set of actions and propose labelling them automatically, so every shift becomes a sequence of actions.
2. **Skate IMUs: how each stride looks.** One IMU on each skate detects when the blade strikes and leaves the ice. This gives contact time and stride time for every stride; adding an IMU on the pelvis makes stride length estimates more accurate [6]. Comparing the two skates gives left/right asymmetry.
3. **Fusion: every stride gets a label.** Each stride is tagged with the action it belongs to and compared with the player's own rested baseline for that action. Single strides are noisy, so values are averaged over many strides before a deviation is flagged.
4. **Classification: fatigue or technique.** A drift that grows during a game in a player who is normally within baseline points to fatigue. A deviation that is present from the start, or in every game, points to poor technique or a strength imbalance. Machine learning has already been used to predict the onset of fatigue from skating kinematics [3].

Why the skates? The skate is the piece of equipment that moves least relative to the body, because it fits snugly around the foot. Sensors on pads or straps shift during play.

Signal processing. A 9-axis IMU sampled at 100–500 Hz captures the blade impacts without draining the battery or overloading storage. A finite state machine splits each stride into glide, push-off and swing phases, and a Kalman filter reduces noise from the ice surface.

Positioning and IMUs have already been combined on real ice, but to recognise shot types rather than to monitor skating [8].

What the coach sees

The app shows the whole team live and flags a player as soon as their strides in one skating action drift outside their own baseline. Each flag names the action, the period and the shift, the size of the deviation, and whether the pattern looks like fatigue or technique. It also suggests what to do: a shorter shift, or a personal drill built from the player's own skating sequences, using the cost-function method of Swarén et al. [7].

Simulated example (player #17, period 3, shift 14, crossover left):

Metric	Value	Rested baseline	Change
Left/right asymmetry	14.1%	3–7%	+7.1 percentage points
Stride length	2.82 m	3.09 m	−9%
Contact time	336 ms	311 ms	+8%

The player's technique holds in periods 1 and 2 and deteriorates in period 3. The RTLS data shows more gliding and lower speed in the same shift. The app classifies the change as fatigue and suggests shorter shifts and a 15-second crossover drill. All values are simulated to illustrate the concept.

Limitations

- **Not validated on real ice.** Skate IMUs have only been tested in a lab, on synthetic ice and forward skating [6]. Game speeds, crossovers, tight turns and stops remain untested.
- **Noise versus signal.** Single-stride errors are of the same size as the asymmetry we want to detect, so the system depends on averaging and a personal baseline.
- **No direct hip measurement.** Skate IMUs measure the feet, not the hip, knee or upper body, where injuries occur. Hip angles are also the least accurate in markerless capture [5].

- **Arena and equipment.** RTLS needs receivers installed in the arena. Sensors mounted on the outside of the boot may not be allowed in games, and metal in the boards can disturb the IMU magnetometers.
- **Ice conditions.** Grooves in worn ice can change the direction of the blade and add variation between strides.
- **Simulated results.** The example values are illustrations, not measurements.

Future work

The next step is to validate the system on real ice, then test whether flagged strides predict injury:

1. **Validate on ice.** Calibrate the skate IMUs against a reference system on real ice, at game speed and for every skating action, not just forward strides.
2. **Define normal asymmetry.** Measure many players to find what normal left/right asymmetry is for each action, and where a stride goes from normal to deviating.
3. **Prospective study.** Group players into low, normal and high asymmetry, link the groups to clinical screening such as hip strength ratios, and follow them over a season for groin, hip and knee pain.
4. **Coach app.** If asymmetry predicts injury, build the warning app for practice and games.

Acting on a risk sign can work. When one NHL team gave a strengthening programme to players screened as having weak adductors, adductor strains fell from 3.2 to 0.71 per 1,000 player-game exposures [9]. Smartphone-based motion capture such as OpenCap could add cheap off-ice checks of hip and knee mechanics [10].

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